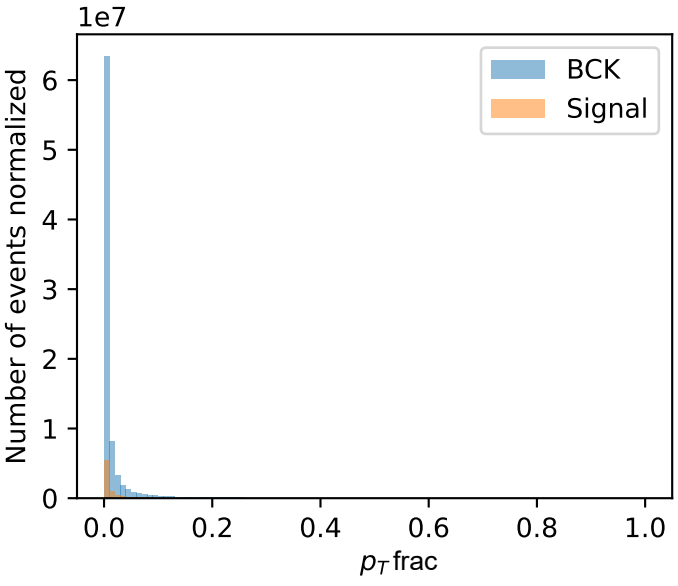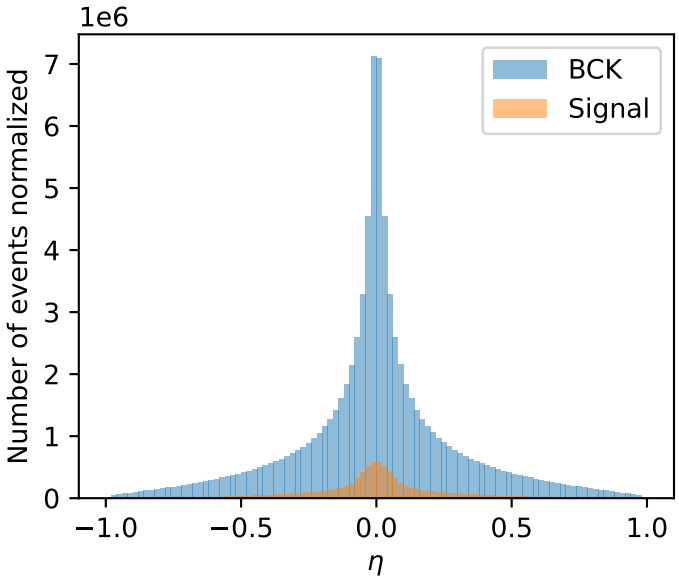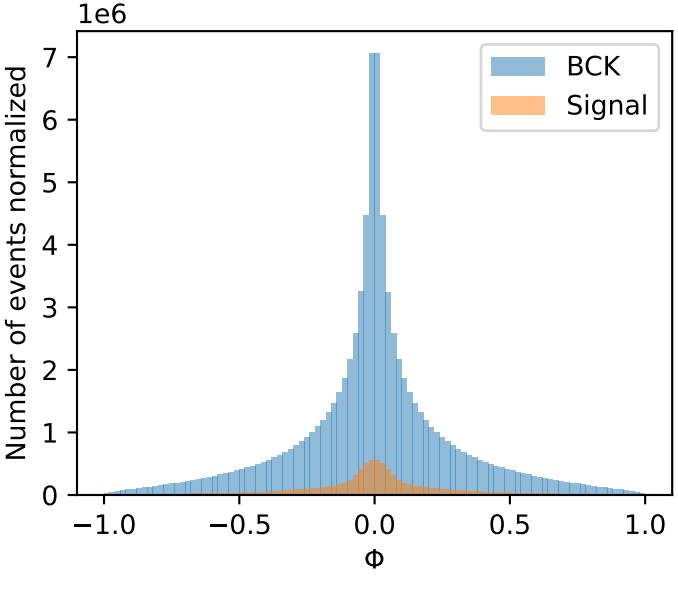Distribution for reconstructed events